

BUSINESS PROPOSAL

Ambulance Booking & Tracking System

Prepared For

Healthcare Organizations, Hospitals, Emergency Service Providers, Government Authorities & Private Ambulance Operators

Prepared By

NewGenXperts Solutions Private Limited

1. Executive Summary

The Ambulance Booking & Tracking System is a smart healthcare mobility platform designed to streamline emergency medical transportation through real-time ambulance booking, intelligent dispatching, GPS-based live tracking, and centralized operational management.

The proposed solution will help hospitals, ambulance operators, healthcare providers, and emergency response teams improve response time, optimize ambulance utilization, reduce operational inefficiencies, and ensure faster medical assistance during emergencies.

The platform will be available on both Mobile and Desktop and will support users, drivers, supervisors, hospitals, and administrators through an integrated digital ecosystem.

2. Project Objectives

The primary objective of this project is to develop a scalable and intelligent ambulance management ecosystem that enables:

- Faster ambulance booking and dispatch
- Real-time ambulance tracking
- Smart allocation and routing
- Emergency response optimization
- Centralized monitoring and control
- Improved healthcare accessibility
- Reduced request rejection and missed emergency cases

3. Proposed Solution

The Ambulance Booking & Tracking System will provide a modern emergency healthcare platform with the following core capabilities:

Core Modules

A. User Application

- User Registration & Login
- Emergency Ambulance Booking
- Live GPS Tracking
- Auto Location Detection
- Estimated Arrival Time
- Facility Selection During Booking
- Payment Integration
- Booking History
- Emergency Contact Sharing

B. Ambulance Driver Application

- Driver Authentication
- Ride Acceptance Workflow
- Live Navigation Support
- Real-Time Status Updates
- Route Optimization
- Availability Management

C. Supervisor Dashboard

- Centralized Request Monitoring
- Ambulance Assignment Management
- No-Rejection Dispatch Mechanism
- Driver & Fleet Monitoring
- Emergency Escalation Management
- Analytics & Reports

D. Admin Panel

- User Management
- Driver Management
- Hospital Management
- Ambulance Fleet Management

- Reports & Analytics
 - Payment & Subscription Monitoring
 - System Configuration
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4. Key Features

4.1 Ambulance Booking System

Users can instantly book ambulances using:

- Current GPS Location
- Manual Pickup Selection
- Hospital Destination Selection
- Emergency Category Selection

4.2 Real-Time Tracking

The platform will provide:

- Live GPS Tracking
- Driver Location Monitoring
- Route Visibility
- Estimated Time of Arrival
- Real-Time Notifications

4.3 Facility-Based Ambulance Selection

Users can select ambulance facilities such as:

- Oxygen Support
- Doctor Availability
- Nurse Availability
- Saline Support
- ICU Ambulance
- Ventilator Support

4.4 Bike Ambulance Integration

The system will support Bike Ambulance services for:

- Faster response in traffic congested areas
- Immediate first aid response
- Rural and narrow road access

4.5 First Medical Assistance Network

The platform will help users identify:

- Nearby Medical Staff
- First Responders
- Nearest Hospitals
- Emergency Healthcare Facilities

4.6 No-Rejection Assignment System

To avoid missing emergency requests:

- Drivers cannot reject emergency requests directly
 - All requests will be routed to Supervisor Control
 - Supervisor assigns available ambulances
 - Ensures operational accountability
 - Minimizes emergency response failure
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5. Technology Stack

Frontend

- React Native (Mobile Application)
- React.js (Web Dashboard)

Backend

- Node.js
- Express.js

Database

- MongoDB

Cloud & Hosting

- AWS / Azure / Google Cloud

Maps & Tracking

- Google Maps API
- GPS Tracking Integration

Real-Time Communication

- Socket.IO
- Push Notifications

6. System Architecture

User Mobile App ↓ API Gateway / Backend Server ↓ Dispatch & Tracking Engine ↓
MongoDB Database ↓ Admin & Supervisor Dashboard ↓ Ambulance Driver Application

7. Business Benefits

For Hospitals

- Faster patient transportation
- Improved emergency management
- Better patient care coordination

For Ambulance Operators

- Improved fleet utilization
- Real-time operational visibility
- Reduced idle time
- Increased booking efficiency

For Patients & Citizens

- Faster emergency response
- Better healthcare accessibility
- Transparent ambulance tracking
- Reliable emergency assistance

For Government & Smart Cities

- Digital healthcare infrastructure
 - Improved emergency response network
 - Data-driven healthcare planning
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8. Security & Compliance

The platform will include:

- Secure User Authentication
- Role-Based Access Control
- Data Encryption

- Secure Cloud Infrastructure
 - API Security
 - Backup & Disaster Recovery
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9. Project Deliverables

The project deliverables will include:

- Mobile Application (Android & iOS)
 - Web Admin Dashboard
 - Supervisor Control Panel
 - Driver Application
 - Backend APIs
 - Database Setup
 - GPS Tracking Integration
 - Deployment & Hosting Support
 - Documentation & Training
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10. Implementation Timeline

Phase	Activities	Duration
Phase 1	Requirement Analysis & Planning	1 Week
Phase 2	UI/UX Design	1-2 Weeks
Phase 3	Backend Development	3-4 Weeks
Phase 4	Mobile & Web Development	4-6 Weeks
Phase 5	GPS & Real-Time Integration	1-2 Weeks
Phase 6	Testing & QA	1-2 Weeks
Phase 7	Deployment & Training	1 Week

Estimated Total Timeline: 10–14 Weeks

11. Commercial Proposal

Estimated Project Cost Breakup

The following is an estimated commercial breakup for the Ambulance Booking & Tracking System project.

Module / Component	Estimated Cost (INR)
Requirement Analysis & Project Planning	₹ 75,000
UI/UX Design (Mobile + Web)	₹ 1,25,000
Mobile App Development (React Native)	₹ 3,50,000
Web Admin Dashboard Development	₹ 2,00,000
Driver Application Development	₹ 1,50,000
Supervisor Dashboard Development	₹ 1,25,000
Backend API Development (Node.js)	₹ 2,50,000
MongoDB Database Design & Integration	₹ 75,000
Google Maps & GPS Tracking Integration	₹ 1,25,000
Real-Time Tracking (Socket.IO)	₹ 1,00,000
Ambulance Booking Engine	₹ 1,25,000
Bike Ambulance Module	₹ 75,000
First Medical Assistance Module	₹ 1,00,000
Hospital & Medical Staff Integration	₹ 1,00,000
Notification System (SMS/Push/Email)	₹ 75,000
Authentication & Security Implementation	₹ 80,000
Payment Gateway Integration	₹ 75,000
Testing & Quality Assurance	₹ 1,00,000
Cloud Deployment & Server Setup	₹ 80,000
Documentation & Training	₹ 50,000
Project Management & Coordination	₹ 1,00,000

Estimated Total Project Cost

₹ 24,10,000 INR

Optional Additional Enhancements

Additional Feature	Estimated Cost
AI-Based Smart Dispatch System	₹ 2,50,000
Live Video Consultation	₹ 1,50,000
IoT Medical Device Integration	₹ 2,00,000
Multi-Language Support	₹ 75,000
Advanced Analytics Dashboard	₹ 1,25,000
Insurance Integration	₹ 1,50,000
Smart City API Integration	₹ 2,00,000

Third-Party / Recurring Costs (Approx.)

Service	Estimated Monthly / Annual Cost
Cloud Hosting (AWS/Azure)	₹ 15,000 – ₹ 50,000 / Month
Google Maps API	Usage Based
SMS Gateway	Usage Based
Push Notification Services	Minimal / Free Tier
Domain & SSL	₹ 5,000 – ₹ 15,000 / Year
Maintenance & Support AMC	15% – 20% of Project Cost / Year

Recommended Development Team

Role	Resource Count
Project Manager	1
UI/UX Designer	1
React Native Developer	2
Backend Developer	2
QA/Test Engineer	1
DevOps/Cloud Engineer	1

Estimated Project Duration

10–14 Weeks depending upon feature scope, integrations, and deployment scale.

12. Future Enhancements

The system can be extended with:

- AI-Based Ambulance Prediction
 - Smart Route Optimization
 - Hospital Bed Availability Integration
 - Emergency Video Consultation
 - IoT Medical Device Integration
 - Multilingual Support
 - Smart City Integration
 - Insurance & Payment Automation
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13. Why NewGenXperts Solutions

NewGenXperts Solutions Private Limited specializes in building scalable digital solutions with expertise in:

- Healthcare Technology
- Mobile & Web Applications
- Real-Time Tracking Systems
- Cloud Solutions
- AI & Smart Automation
- Enterprise Software Development

Key Strengths:

- Experienced Development Team
 - Agile Development Methodology
 - End-to-End Product Delivery
 - Scalable Architecture Design
 - Dedicated Technical Support
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14. Conclusion

The Ambulance Booking & Tracking System is designed to modernize emergency healthcare transportation through technology-driven automation, real-time monitoring, and intelligent dispatching.

The proposed platform will significantly improve emergency response efficiency, enhance patient safety, and create a scalable digital healthcare ecosystem for hospitals, ambulance providers, and smart city initiatives.

We look forward to collaborating and delivering a robust healthcare mobility solution.

Contact Information

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